

Hair Follicle On-a-Chip

to investigate the effects of Minoxidil on androgenetic alopecia

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ABSTRACT

The investigation of androgenetic alopecia (AGA) is hindered by the lack of human-relevant in vitro models capable of reproducing androgen-driven hair follicle miniaturization under controlled conditions. Although Minoxidil is the most widely used treatment, its preclinical evaluation is limited by the absence of platforms that enable precise pharmacological control and functional readouts. In this work, we present a commercial microfluidic hair follicle-on-a-chip designed to enable controlled modeling of AGA and therapeutic screening with Minoxidil in vitro.

The device is a single-organ-on-chip fabricated from cyclo olefin copolymer (COC), polyethylene terephthalate (PET), and bioPET 002. A spatially organized three-dimensional construct integrates dermal papilla spheroids, outer root sheath keratinocytes, and dermal fibroblasts embedded in a collagen matrix, enabling sustained epithelial–mesenchymal interactions under air–liquid interface conditions.

An open circuit perfusion strategy allows stable and precise coexposure to dihydrotestosterone and Minoxidil, while follicular responses are monitored noninvasively through fluorescence-based quantification of Wnt/ β -catenin activity and trichoscopic assessment of follicular morphology. By combining controlled drug delivery, physiologically relevant cell organization, and real time functional readouts, this platform provides a reproducible, scalable tool for mechanistic studies and preclinical screening of Minoxidil under androgenetic alopecia like conditions.

STATE-OF-THE-ART

AGA is a multifactorial condition characterized by a genetically determined hypersensitivity to androgens, leading to progressive hair follicle miniaturization. Genetic variants increase androgen receptor density or sensitivity in susceptible scalp regions, where elevated dihydrotestosterone levels and enhanced 5 α -reductase type II activity shorten the anagen phase, prolong telogen, and reduce the anagen/telogen ratio^[1]. Minoxidil remains the most widely used treatment for AGA as it promotes hair follicle growth by enhancing dermal papilla activity and modulating signaling pathways regulating follicular activation. These effects include stimulation of growth-supportive cues and reactivation of canonical Wnt/ β -catenin signaling, promoting matrix cell proliferation and prolongation of the anagen phase^[2].

Recent advances in vitro modeling aim to improve the study of follicular physiology and drug responses. In this context, a group of scientists from the Research and Innovation Center (Yongin-si, Republic of Korea) developed human hair follicle organoids that recapitulate key aspects of follicular architecture; however, their approach relies on scarce human follicle-derived cells and exhibits limited long-term growth and maturation^[3]. On the other hand, in the Yokohama National University (Japan), researchers reported a microfluidic platform enabling large-scale generation of hair follicle germ aggregates under controlled microenvironments, but these constructs lack full epithelial-mesenchymal organization and fail to consistently produce mature hair shafts^[4]. In parallel, an European research group reviewed perfusion and multicellular support^[5]; nevertheless, most commercial and experimental systems still lack functional human hair follicles.

Collectively, these limitations highlight the need for advanced, reproducible human follicle-on-chip platforms capable of evaluating therapeutic compounds under physiologically relevant conditions.

AIMS, METHODOLOGY AND FORESEEN RESULTS

The aim of this project is to design a microfluidic hair follicle-on-a-chip capable of reproducing key structural and functional features of the human hair follicle and modeling AGA. This OoC will simulate DHT-induced follicular miniaturization, and enable reproducible drug-response testing to assess the therapeutic efficacy of Minoxidil.

As they are hair follicles that are being modelled, our device is a single-organ-on-chip, enabling a high degree of biological authenticity. Furthermore, we will proceed with a bottom-up approach due to the difficulty of obtaining biopsies for every experiment in the case we would have followed an organotypic approach.

Materials

As materials we will use COC due to its low absorption of hydrophobic drugs (as Minoxidil is), allowing us to work with reliable concentration gradients and obtain true dose-response curves. This material offers excellent optical transparency, allowing real-time imaging of the follicle^[6]. On the other hand, PET was chosen because it shows minimal drug absorption^[7], stable and reproducible diffusion properties^[8], and excellent compatibility with endothelial and epithelial co-culture^[9], making it ideal for reliable Minoxidil screening in our chip. This material is also excellent for imaging, microscopy, and general cell culture due to optical clarity and good cell attachment when treated^[10]. Moreover, in an attempt to reduce the environmental footprint of our device we will use BioPET 002, a partially bio-based polyethylene terephthalate. Both materials can be sterilized with ETO and UV irradiation^[11].

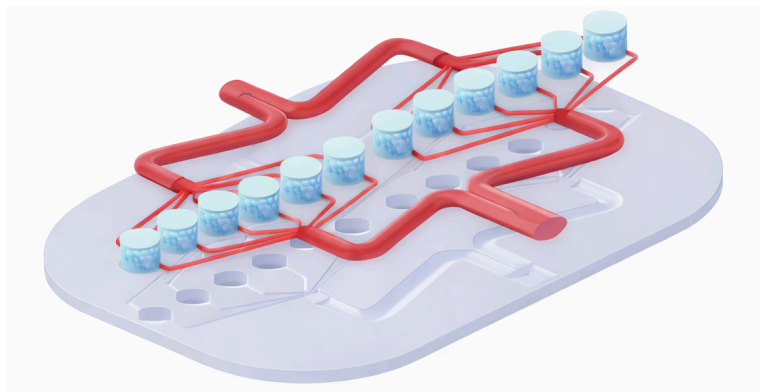


Figure 1. 3D design of the Hair Follicle-on-a-Chip integrating dynamic perfusion and cell culture compartments. Developed using Blender.

Cells and 3D follicular architecture

In this hair follicle-on-a-chip model we combine three human cell populations. Immortalized dermal papilla cells (IDP) are included as the inductive mesenchymal core responsible for hair follicle signaling and for mediating both androgen suppression and Minoxidil-driven rescue effects^{[13][15]}. ORS keratinocytes constitute the epithelial compartment, enabling follicular structure formation, outer root sheath organization, and keratin expression relevant to hair-shaft production^[12]. Finally, dermal fibroblasts are embedded throughout a collagen matrix to provide extracellular-matrix remodeling, mechanical support, and paracrine factors required for follicle viability and cycling.

To generate a hair follicle-like three-dimensional construct, cells were organized in a spatially controlled manner to reproduce the main compartments of the hair-bearing scalp^[14].

The dermal compartment was established by embedding dermal fibroblasts within a collagen type I matrix. Collagen type I provides mechanical support and enables fibroblast-mediated matrix contraction and remodeling, closely resembling the native dermal extracellular matrix^[14,16]. Fibroblast-contracted collagen gels have been shown to support epithelial-mesenchymal interactions (EMIs) required for folliculoid organization^[14].

In parallel, immortalized dermal papilla cells were aggregated into three-dimensional spheroids. It is demonstrated that DP three-dimensional aggregation partially restores their gene expression and functional properties^[15,3]. Once stabilized, DP spheroids were incorporated into the collagen-fibroblast matrix, positioning them within the dermal compartment to enable epithelial-mesenchymal signaling^[14,18].

After matrix stabilization, ORS keratinocytes were seeded onto the surface of the collagen construct. The culture was initially maintained under fully submerged conditions on DMEM/f12 supplemented with 10% FBS to promote keratinocyte attachment, spreading, and formation of a continuous epithelial layer, as it provides a balanced mix of nutrients^[16,20].

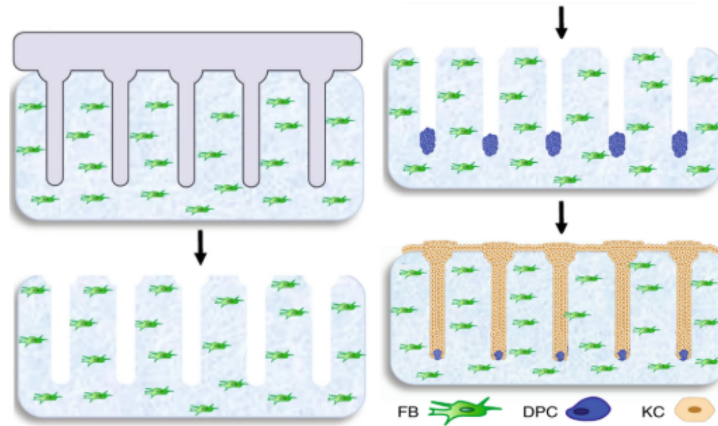


Figure 2. Array of microwells to form spontaneous aggregates of dermal papilla cell using moulds^[28].

Following culture stabilization, it was transitioned to an air–liquid interface (ALI) by reducing the medium volume in the apical compartment.

The experiment will be divided into three stages, each of them employing an open blood circuit which prevents the accumulation of waste products and allows precise control of Minoxidil and hormone exposure over time. The *in vitro* growth period of hair follicles is recommended to be extended to 14 days^[21]. The effects of DHT on follicle growth and signaling can be documented over up to 10 days of continuous exposure, making this a relevant timeframe for in-chip induction of androgenetic changes^[19]. The duration of the Minoxidil treatment can be determined by each laboratory according to its experimental design.

Monitorization and expected results

Device monitoring is addressed using fluorescence microscopy (FM) and trichoscopy (TC).

FM is employed to non-invasively monitor Wnt/ β -catenin signaling dynamics in real time. Hair follicle cells express a LEF/TCF-responsive yellow fluorescent protein (YFP) reporter, whose expression increases upon Wnt activation, together with a constitutively expressed cyan fluorescent protein (CFP) used as an internal reference to control for cell number and expression variability. The YFP/CFP fluorescence ratio is used as a quantitative proxy for pathway activation over time.

During the first stage, corresponding to normal follicle growth and maintenance of an anagen-like state, a progressive increase in YFP fluorescence is expected, particularly in matrix and progenitor-rich regions, reflecting robust Wnt/ β -catenin activity. CFP fluorescence remains stable, resulting in an increased and stabilized YFP/CFP ratio indicative of a healthy, growing follicle.

In the second stage, alopecia is induced, leading to a marked reduction in YFP fluorescence due to suppression of LEF/TCF-mediated transcription and canonical Wnt signaling. CFP levels remain unchanged, and the resulting decrease in the YFP/CFP ratio reflects follicular miniaturization and transition toward a catagen- or telogen-like state.

In the third stage, following drug treatment, YFP fluorescence is expected to recover. Minoxidil promotes Wnt/ β -catenin signaling by enhancing dermal papilla function and follicular microenvironmental support, leading to a progressive increase in the YFP/CFP ratio that reflects signaling reactivation in responsive follicles^[22,23,24,25].

On the other hand, TC is a non-invasive diagnostic method that allows improved visualization of morphological structures not visible to the naked eye^[26]. In androgenetic alopecia, the number of yellow dots, pilosebaceous units containing only a single hair, and units with perifollicular hyperpigmentation is significantly increased ($p < 0.05$). Additionally, the percentage of thin hairs (< 0.03 mm) in AGA is significantly higher than in healthy controls^[27].

RELEVANCE OF THE PROJECT WITH RESPECT OF THE STATE OF THE ART

This project proposes a hair follicle-on-a-chip platform specifically optimized for the preclinical investigation of Minoxidil, addressing critical limitations that currently hinder its mechanistic study and translational relevance. Although Minoxidil is the most widely prescribed treatment for androgenetic alopecia, its efficacy depends on tightly controlled concentration gradients, vascular delivery, and epithelial-mesenchymal signaling, which are poorly reproduced in existing in vitro models.

Current follicle cultures and organoid systems are mostly static^[4], rely on donor-limited follicle-derived cells^[3], and lack controlled pharmacological exposure^[4]. Moreover, many microfluidics platforms employ PDMS, whose high absorption of hydrophobic drugs compromises accurate Minoxidil testing^[4]. In contrast, this platform uses thermoplastics (COC and PET) with minimal drug sorption, enabling precise and reproducible control of Minoxidil concentrations and reliable dose-response assessment.

A major innovation is the integration of an open-circuit, one-pass perfusion strategy that mimics systemic hormone delivery conditions and continuous removal of Minoxidil from the culture. Continuous perfusion stabilizes hormone and drug levels, prevents metabolite accumulation, and enables controlled co-exposure to dihydrotestosterone and Minoxidil, reproducing clinically relevant competitive signaling environments observed in AGA^[2].

Biologically, the system combines IDP spheroids, ORS keratinocytes, and dermal fibroblasts in a spatially organized three-dimensional construct under an ALI. This configuration supports sustained epithelial-mesenchymal interactions and enables simultaneous investigation of topical application, vascular-like nutrient supply, follicular morphology, proliferation, and growth factor expression^[14,16,18,20].

Overall, this hair follicle-on-a-chip overcomes key limitations of current organoid and microfluidic approaches, providing a reproducible, scalable, and physiologically relevant platform uniquely suited for Minoxidil screening and mechanistic studies under androgenetic alopecia-like conditions.

BUDGET

Our proposed budget is €12.258,96 and it is based on the costs associated with cell culture reagents and consumables, drugs and hormone treatments, fabrication and operation of the microfluidic platform, and experimental monitoring and analysis.

MATERIAL	COST	COMPANY
CELL CULTURE		
Immortalized Human Hair Follicle Dermal Papilla Cells	€5.837,47	Creative Bioarray
Immortalized Human Follicle Outer Root Sheath Cells	€4.173	Topsan
Immortalized Human Dermal Fibroblast	€623,63	Cells Online LLC
DMEM/F-12 (with FBS 10%), (SLM-243-B)	€272,73	Sigma/Merk
Collagen type I	€365	Sigma/Aldrich
DRUGS/HORMONES		
Dihydrotestosterone (DHT)	€42,38	GLPBio
MICROFLUIDIC PLATFORM / FABRICATION		
Thermoplastic COC resin	€474	Sigma/Aldrich
PET membrane (HTTP01300)	€110,75	Sigma/Aldrich
BioPET 002	€210	NaturePlast

MONITORING		
YFP	€75	ADDGENE
CFP	€75	ADDGENE
TOTAL COST= €12.258,96		

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